# Microsoft DSA Interview Questions

Commonly Reported Microsoft Data Structures & Algorithms Questions

## Arrays & Strings

1. Two Sum
2. Best Time to Buy and Sell Stock
3. Maximum Subarray (Kadane's Algorithm)
4. Product of Array Except Self
5. Merge Intervals
6. Rotate Array
7. Move Zeroes
8. Longest Consecutive Sequence
9. Container With Most Water
10. Trapping Rain Water
11. Longest Substring Without Repeating Characters
12. Longest Palindromic Substring
13. String Compression
14. Valid Anagram
15. Group Anagrams
16. Minimum Window Substring

## Linked Lists

1. Reverse Linked List
2. Merge Two Sorted Lists
3. Detect Cycle in Linked List
4. Intersection of Two Linked Lists
5. Remove Nth Node From End
6. Copy List With Random Pointer
7. LRU Cache
8. Flatten Multilevel Linked List

## Stacks & Queues

1. Valid Parentheses
2. Min Stack
3. Implement Queue Using Stacks
4. Next Greater Element
5. Largest Rectangle in Histogram

6. Sliding Window Maximum
7. Daily Temperatures

## Trees & BST

1. Binary Tree Level Order Traversal
2. Lowest Common Ancestor
3. Validate BST
4. Kth Smallest Element in BST
5. Diameter of Binary Tree
6. Serialize and Deserialize Binary Tree
7. Binary Tree Zigzag Traversal
8. Path Sum
9. Maximum Path Sum
10. Vertical Order Traversal

## Graphs

1. Number of Islands
2. Clone Graph
3. Course Schedule
4. Pacific Atlantic Water Flow
5. Word Ladder
6. Graph Valid Tree
7. Dijkstra's Algorithm
8. Shortest Path in Binary Matrix
9. Alien Dictionary
10. Network Delay Time

## Dynamic Programming

1. Climbing Stairs
2. House Robber I & II
3. Coin Change
4. Longest Increasing Subsequence
5. Longest Common Subsequence
6. Edit Distance
7. Knapsack Problem
8. Partition Equal Subset Sum
9. Decode Ways
10. Word Break

## Recursion & Backtracking

1. Subsets
2. Permutations
3. Combination Sum
4. N-Queens
5. Sudoku Solver
6. Generate Parentheses
7. Letter Combinations of Phone Number

## Heaps & Priority Queues

1. Kth Largest Element
2. Top K Frequent Elements
3. Merge K Sorted Lists
4. Find Median From Data Stream
5. Meeting Rooms II

## Binary Search

1. Search in Rotated Sorted Array
2. Find Peak Element
3. Median of Two Sorted Arrays
4. First and Last Position in Sorted Array
5. Koko Eating Bananas

## Bit Manipulation

1. Single Number
2. Missing Number
3. Counting Bits
4. Reverse Bits
5. Power of Two

## Advanced Microsoft-Specific Questions

1. Design a File System
2. Design a Search Autocomplete System
3. Design Excel Formula Engine
4. Design URL Shortener
5. Design Microsoft Teams Chat Service

6. Design Distributed Cache
7. Design OneDrive Storage System